

INDIAN INSTITUTE OF TECHNOLOGY,
HYDERABAD

BASIC BIOINFORMATICS

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ASSIGNMENT 5

Dotplot

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1.a Repeated Domains

Repeated domains refer to segments within a protein sequence where similar structural units occur multiple times in succession. These repetitions usually arise due to duplication events and often contribute to similar functional roles within the protein.

In a dot plot, repeated domains are observed as multiple diagonal lines running parallel to the principal diagonal. This pattern indicates that certain regions of the sequence are duplicated at different positions.

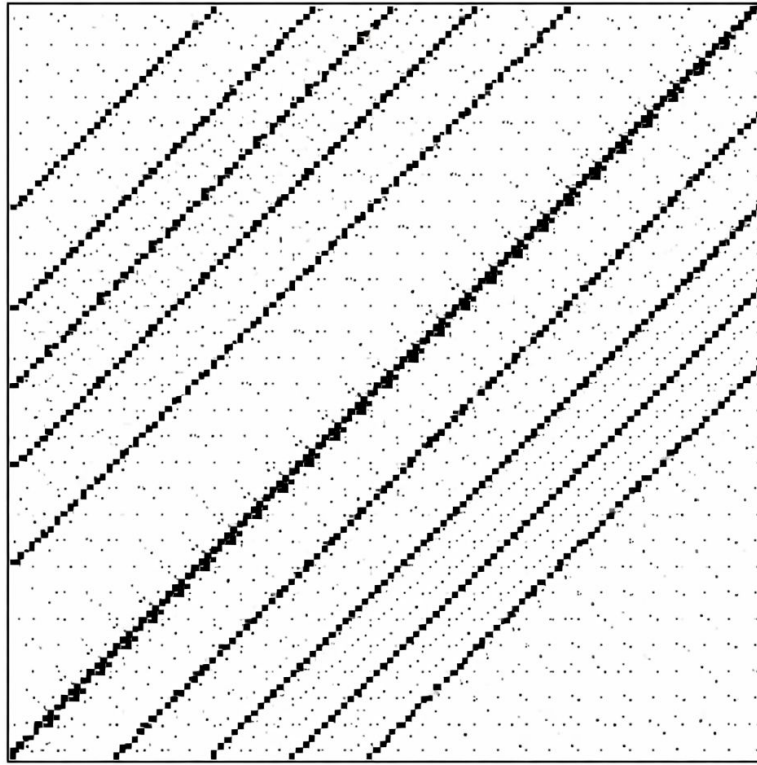


Figure 1: Visualization of repeated domains in a dot plot

1.b Conserved Domains

Conserved domains are specific regions of a protein that retain similar structure and function across different organisms due to evolutionary pressure. These domains are typically associated with essential biological functions.

In dot plot analysis, conserved domains appear as continuous or strong diagonal alignments. These diagonals reflect regions that remain highly similar between sequences, often corresponding to functional or structural motifs that are preserved across species.

These regions indicate high sequence similarity, reflecting strong evolutionary conservation across species.

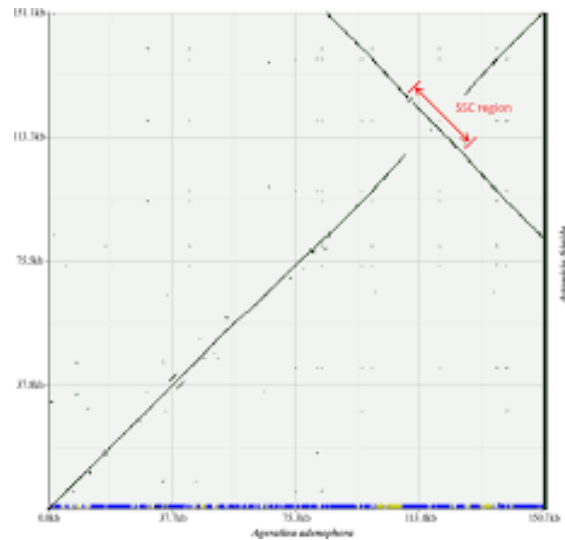


Figure 2: Conserved domain representation in a dot plot

1.c Exons and Introns

Exons are coding segments of a gene that are retained in the mature RNA and translated into proteins, whereas introns are non-coding segments that are removed during RNA processing.

In dot plots, exon regions typically appear as clear diagonal segments since they

align directly between genomic DNA and processed RNA sequences.

Introns, on the other hand, are represented as interruptions or gaps in the diagonal pattern because they are absent in the processed sequence, leading to discontinuities.

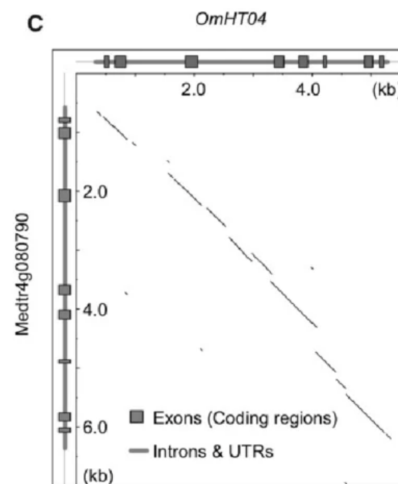


Figure 3: Dot plot showing exon and intron regions

The dot plot shows diagonal segments corresponding to exon regions, where alignment between genomic DNA and processed RNA is preserved. These segments are separated by distinct gaps, which represent intron regions removed during RNA splicing. The discontinuous diagonal pattern clearly differentiates coding and non-coding regions.

1.d Frameshifts

A frameshift mutation occurs when nucleotides are inserted or deleted in numbers not divisible by three, thereby altering the reading frame of the sequence.

In a dot plot, frameshifts are visualized as abrupt shifts or breaks in the diagonal alignment. Instead of a smooth diagonal, the alignment appears offset, indicating that the sequence has been disrupted due to an insertion or deletion event.

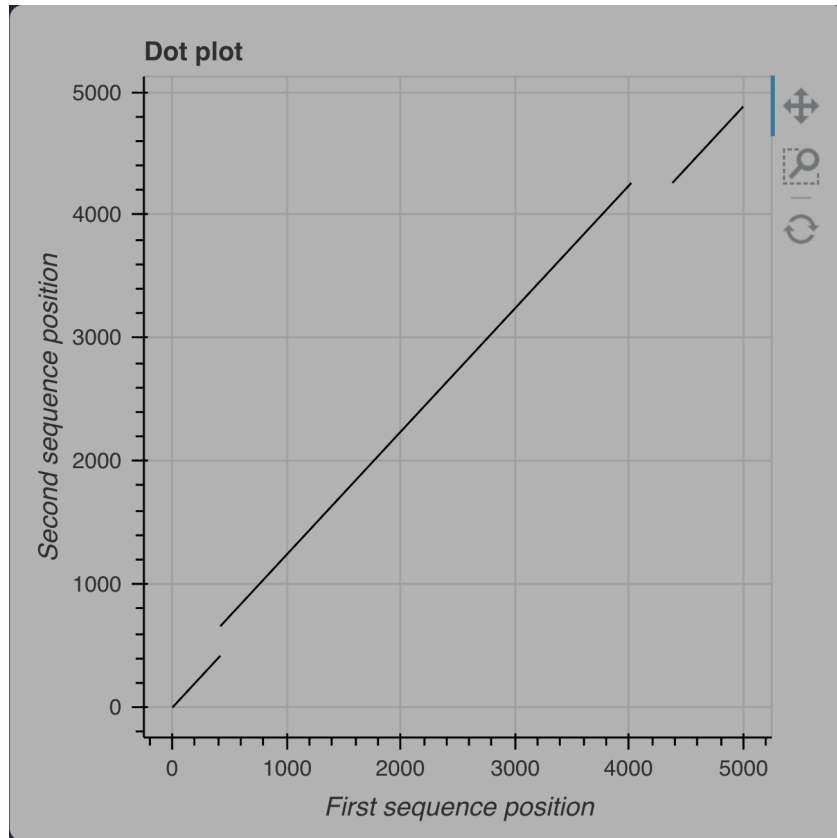


Figure 4: Effect of a frameshift mutation on a dot plot

The dot plot shows a disruption in the continuous diagonal alignment, followed by a shifted continuation. This indicates the presence of an insertion or deletion event that alters the reading frame, resulting in a frameshift. The discontinuities in the diagonal further support the presence of sequence misalignment after the mutation.

2 Region Identification of *Drosophila melanogaster*

The self-comparison dot plot for *Drosophila melanogaster* was generated by retrieving the protein sequence in FASTA format from the NCBI database and analyzing it using the EMBOSS Dotmatcher tool.

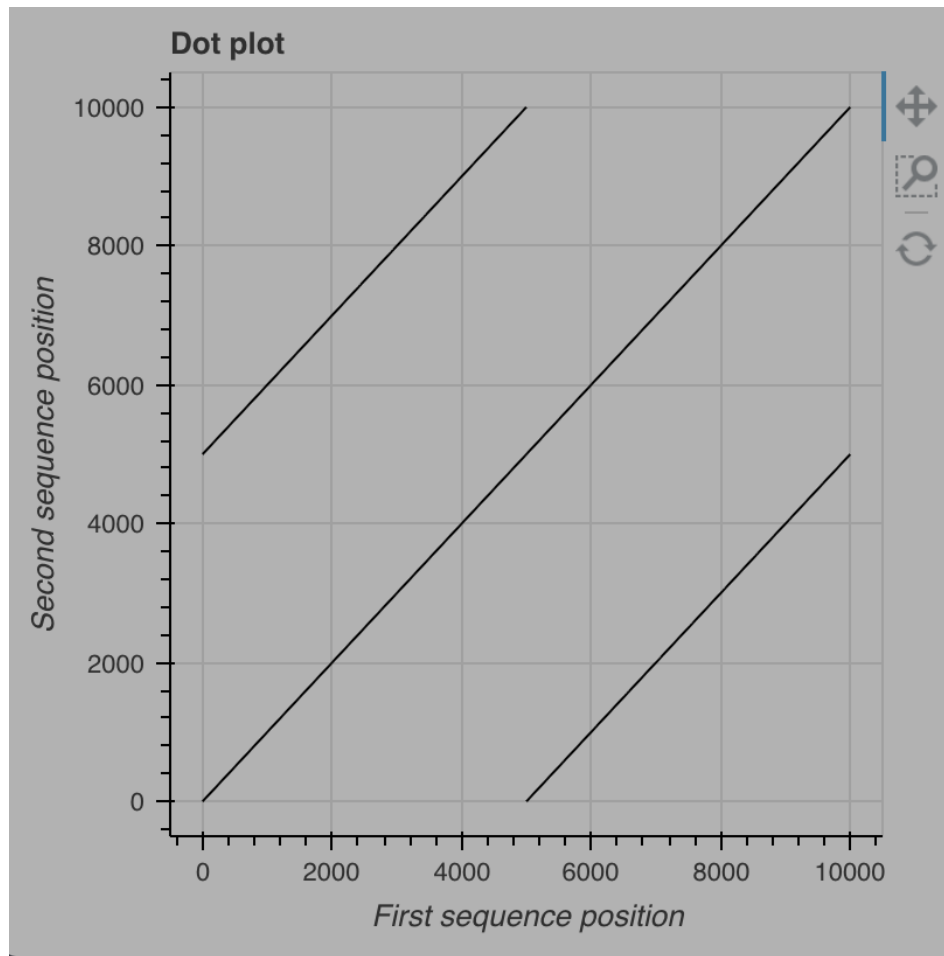


Figure 5: Self-alignment dot plot of *Drosophila melanogaster*

- **Primary Diagonal (Self Similarity):** A prominent diagonal extending from the origin to the end of the plot indicates complete alignment of the sequence with itself.
- **Repeated Regions:** Multiple diagonal lines parallel to the main axis are observed, indicating the presence of repeating sequence motifs arranged in tandem within the protein.
- **Unique Region (1400–1500):** The final portion displays only the main diagonal without additional parallel lines, implying that this segment does not contain repetitive elements.

3 Comparison of Insulin Sequences Across Species

The insulin protein sequences from human, pig, and rat were compared using alignment tools such as BLAST and Clustal Omega. Multiple scoring matrices were used to evaluate similarity.

Table 1: Alignment Results: Human vs Rat Insulin

Matrix	Score	Identity (%)	E-value
BLOSUM62	171	82.73%	1×10^{-61}
PAM250	151	82.73%	3×10^{-57}
PAM30	178	82.73%	6×10^{-56}

Table 2: Alignment Results: Human vs Pig Insulin

Matrix	Score	Identity (%)	E-value
BLOSUM62	131	85.45%	7×10^{-46}
PAM250	112	83.64%	5×10^{-41}
PAM30	139	83.64%	1×10^{-41}

Observations

- Human and pig insulin sequences show very high similarity, indicating strong evolutionary conservation.
- Rat insulin is slightly more divergent but still retains key functional regions.
- Different scoring matrices affect alignment scores but not the percentage identity significantly.

4 Comparison of Human Hexokinase-1 and Glucokinase

The protein sequences of Hexokinase-1 (HK1) and Glucokinase (HK4) were analyzed using BLASTp with different substitution matrices to compare their similarity.

Table 3: Alignment Results for HK1 vs Glucokinase

Matrix	Max Score	Total Score	Identity (%)	E-value
BLOSUM62	510	983	54.34%	1×10^{-177}
PAM250	456	885	54.34%	3×10^{-169}

Interpretation

- **Strong Homology:** Extremely low E-values confirm that both proteins share a common evolutionary origin.
- **Matrix Dependence:** Although identity remains constant, alignment scores vary depending on the substitution matrix used.
- **Protein Length Difference:** The higher total score is due to Hexokinase-1 being significantly longer, allowing multiple alignment regions with Glucokinase.